TEXORA · COURSE NOTES · LECTURE SHEET

Surface Tension

Why liquid surfaces behave like stretched skins — the molecular picture, surface energy, excess pressure in drops and bubbles, contact angle, and capillarity.

PHY-1103 · PHYSICS-I · TOPIC 3 · LEVEL 1 TERM 1

1. The molecular picture

A molecule deep inside a liquid is pulled equally in all directions by its neighbours; the forces cancel. A molecule at the **surface** has neighbours only below and beside it — the pull from the liquid side is unopposed, so the surface layer is dragged inward, and the surface contracts to the smallest possible area, exactly as a stretched membrane would. Every familiar sign follows: droplets pull themselves spherical (the smallest area for a given volume), a needle can rest on water in a tiny dent of this "skin", insects walk on ponds, and liquid threads break up into beads. **Surface tension** T quantifies the effect in either of two equivalent dresses:

- As a **force**: the force acting per unit length along any line drawn in the surface, pulling perpendicular to the line and tangent to the surface — T in N m^{-1}.
- As an **energy**: the work needed to increase the surface area by one unit — surface energy per unit area, in J m^{-2}. (N m^{-1} and J m^{-2} are the same unit wearing two hats, and the equivalence is a standard examination step.)

Reference values at room temperature: water ≈ 0.073 N m^{-1}, mercury ≈ 0.47 N m^{-1} (unusually high — hence its tight, non-wetting beads), soap solution ≈ 0.03 N m^{-1}, most oils 0.02–0.04 N m^{-1}.

2. Surface energy

Enlarging a surface means dragging molecules from the bulk into the surface against the inward pull — work must be done, and it is stored as surface energy, U = T × (new area created). A soap film on a frame of width l, pulled out by dx, illustrates the bookkeeping: the film has *two* surfaces, the surface force is 2Tl, and the work 2Tl dx equals T times the total new area 2l dx — force picture and energy picture agreeing exactly. When a drop divides into droplets (spraying, atomising), the total area grows enormously and the work of division is why atomisers need pressure — and why fine mists, once made, carry so much surface physics in them.

3. Excess pressure inside drops and bubbles

A curved surface's inward pull must be balanced by a higher pressure inside the curve. For a spherical droplet of radius R, balancing the surface force around a hemisphere against the pressure force gives the **excess pressure**

$$\Delta p = 2T / R \quad \text{(droplet — one surface)} \qquad \Delta p = 4T / R \quad \text{(soap bubble — two surfaces)}$$

The 1/R law has real teeth: the smaller the drop, the higher its internal pressure — which is why small droplets evaporate and feed big ones, and why the tiniest bubbles in a foam collapse first. Worked scale: a water droplet of radius 1.0 mm has $\Delta p = 2 \times 0.0728 / 0.001 \approx$ **146 Pa**; shrink the radius to 1 μm and the excess is a hundred times larger, about 1.46×10^5 Pa — comparable to atmospheric pressure itself.

4. Angle of contact and capillarity

Where a liquid surface meets a solid, it meets it at the **angle of contact** θ, measured through the liquid between the solid surface and the tangent to the liquid surface at the contact line. $\theta < 90°$ means the liquid **wets** the solid (water on clean glass, $\theta \approx 0°$); $\theta > 90°$ means it does not (mercury on glass, $\theta \approx 140°$; water on a waxed or silicone-finished surface). Wetting is decided by the contest between liquid–solid attraction (adhesion) and liquid–liquid attraction (cohesion). In a narrow tube (a **capillary**), the meniscus's surface tension pulls the column up (or pushes it down) until its weight balances:

$$h = 2T \cos \theta / (r \rho g)$$

with r the tube radius and ρ the liquid density. Water in a tube of radius 0.50 mm: $h = 2 \times 0.0728 \times 1 / (0.0005 \times 1000 \times 9.8) \approx 0.0297$ m ≈ **3.0 cm** — the working principle of the capillary-tube measurement of the sessional course, and (in fibrous form) of every absorbent fabric. Mercury, with cos θ negative, is *depressed* in a capillary instead. Capillary rise falls as 1/r: wide tubes show almost none, fine pores enormous pull — Jurin's law.

5. What changes surface tension

- **Temperature:** T falls steadily as temperature rises (agitation weakens the net inward pull) and vanishes at the critical temperature, where surface and bulk become indistinguishable.
- **Contamination and surfactants:** substances that gather at the surface — soaps, detergents, wetting agents, oils — lower T markedly, even in small amounts. This is not a side effect; it is the entire design principle of detergents and textile wetting agents.
- **Dissolved salts** (inorganic electrolytes) typically raise the surface tension of water slightly.

Measurement methods met in this course and its sessional: capillary rise ($h \propto T$ by the formula above), the drop-weight/stalagmometer method (a falling drop's weight is held up by surface tension around the tip), Jaeger's method (pressure to force a bubble from a capillary), and Quincke's method for mercury's large drop — the sessional's own experiment.

6. The textile thread: wetting is destiny

Few topics decide more factory outcomes. A fibre assembly is a bundle of capillaries, and whether a dye liquor, scouring bath or finish enters it at all is decided here: cotton, hydrophilic, shows a small contact angle and wicks liquor along its fibres by capillarity (the absorbency test is Jurin's law on a fabric strip); polyester, hydrophobic, shows $\theta > 90°$ to water and must be coaxed — hence **wetting agents** in every recipe, whose job is precisely to cut T and θ until penetration is fast and level. Run the same physics backwards and you get **water-repellent finishes**: coat the fibre so θ rises past 90° and rain beads and rolls off, while vapour still passes. Foam dyeing, spray application, stain release, soil hiding — all are surface-tension engineering, and the engineer who has understood this sheet can read a wetting problem on sight.

7. Glossary — key terms of this topic

TERM	MEANING
Surface tension (T)	Force per unit length along a line in a liquid surface; equivalently surface energy per unit area. $N\ m^{-1} = J\ m^{-2}$.
Surface energy	Work stored in creating unit area of new surface; $U = T \times$ area.
Excess pressure	The extra pressure inside a curved surface: $2T/R$ for a drop, $4T/R$ for a soap bubble.
Angle of contact (θ)	Angle, through the liquid, between the solid surface and the tangent to the liquid surface at contact; below 90° = wetting.
Capillarity	Rise or depression of a liquid in a narrow tube or pore: $h = 2T\cos\theta/(r\rho g)$.
Cohesion / adhesion	Attraction between like molecules / between unlike substances; their contest decides wetting.
Surfactant	A surface-active substance that collects at interfaces and lowers surface tension.

8. Key points and summary

- Surface tension arises from the unbalanced inward pull on surface molecules; the surface minimises its area, and T measures both the force per unit length and the energy per unit area.
- Curved surfaces carry excess pressure 2T/R (drop) and 4T/R (bubble) — fiercer the smaller the radius.
- Contact angle decides wetting; capillarity $h = 2T\cos\theta/(r\rho g)$ drives liquids into tubes and fabrics; temperature lowers T, surfactants lower it dramatically.

9. Practice

Answer in your own words or with full working; brief model answers follow. ($T_{water} = 0.0728$ N m^{-1}, $\rho_{water} = 1000$ kg m^{-3}, $g = 9.8$ m s^{-2}.)

1. Show that N m^{-1} and J m^{-2} are the same unit, and state in one sentence why the two definitions of T must agree.
2. Find the excess pressure inside a soap bubble of radius 2.0 cm formed from a solution of $T = 0.030$ N m^{-1}.
3. Water rises in a capillary of radius 0.25 mm. Find the height, and state what happens to the height if the tube is cut shorter than that height.
4. Mercury is depressed rather than raised in a glass capillary. Connect this to its contact angle and to $\cos\theta$ in the capillary formula.
5. A steel needle floats on water though steel is denser than water. Explain with the surface-tension picture (and why a drop of detergent ends the trick).
6. A fabric strip dipped in water wicks the water up 6 cm; a chemically finished version of the same fabric wicks only 1 cm. Give the surface-physics explanation.
7. Why must a wetting agent be added to a dye bath for polyester, in terms of T and θ?
8. Two droplets of radii 1 μm and 10 μm sit in the same mist. Which has the higher internal pressure, by what factor, and what slow change does this drive in the mist?

Answer key

1. J m^{-2} = (N·m)/m² = N/m. The force definition integrated over a displacement is the work of creating new surface — the energy definition — so the numbers must coincide.
2. $\Delta p = 4T/R = 4 \times 0.030 / 0.020 = 6.0$ Pa.
3. $h = 2 \times 0.0728 / (0.00025 \times 1000 \times 9.8) = 0.1456 / 2.45 \approx 0.059$ m ≈ 5.9 cm. If cut shorter, the water rises to the top and stays there (the meniscus adjusts its curvature) — it does not fountain out.
4. Mercury on glass has $\theta \approx 140° > 90°$, so $\cos\theta$ is negative and h comes out negative — a depression. Mercury coheres to itself more than it adheres to glass.

5. The needle rests in a dent of the stretched surface; the surface tension along the dent's edges has upward components that carry its weight. Detergent lowers T (and wets the steel), the skin weakens, and the needle sinks.
6. The finish has raised the contact angle (made the fibres less wettable), so cos θ — the driving factor in $h = 2T\cos\theta/(r\rho g)$ for the fabric's pores — has fallen; capillary rise falls with it.
7. Polyester is hydrophobic: θ for water is above 90°, so the bath does not spread or penetrate. The wetting agent lowers the liquor's surface tension and the contact angle below 90°, letting the bath wet the fibre and enter the yarn.
8. The 1 μm droplet: $\Delta p \propto 1/R$, so its excess pressure is 10 times the other's. Small droplets therefore evaporate preferentially and the vapour feeds the large ones — the mist coarsens (Ostwald ripening of the droplet population).
